

SELECTING AN EMS PARTNER

Parameter	Remarks
Overall Manufacturing Process	EMS partner should document the overall manufacturing process with proper entry and exit points as well as criteria for entry and exit. Without this product manufacturing cannot be done independently. Many times, only ISO documentation is done, which does not capture the details of the processes followed, especially inputs to each of the process.
Component Engineering	Good EMS partners have a very good component engineering process covering electronics, plastics, sheet metal, etc, so that the supply chain is robust and ensure quality components. In addition to these two, electronic component obsolescence is a big issue, so EMS partners should have a robust obsolescence management process to take care.
Quality Certification and Approvals	There are 3 types of quality certifications that an EMS partner should possess: • Standard ISO 9001 certification to ensure data is captured and the process is consistent and repeatable. • Environmental certification for manufacturing like RoHS, REACH, and other compliant standards related to workers' safety and wellbeing (like OSHA). • Domain-specific certifications like ISO 13485 (medical) AS9100 (avionics) and ISO 16949 (automotive) should be available, if the products fall in these domains. If the partner does not have any certification it is a business risk.
ECN Process	This process has two components: • The impact on the design when there is a change. • The cost of change and how it will impact the manufacturing. Unless this process is understood, ECNs can stop the products moving out of the production line.
New Product Introduction (NPI)	NPI is a process of a new product being introduced to the manufacturing line. NPI needs intense collaboration between the manufacturing team and OEM's engineering team. A well-defined NPI process, from both sides, is a must.
Design Support	When a product has matured, normally R&D support may not be available from the OEM side. However, due to batch variations, design support is always needed for quick resolution of problems. But smaller EMS partners may not have this facility. In such a case, OEMs have to plan for this upfront or sensitise the EMS partner for the same.
Testing and Tester Design	When the product's volumes are low, expensive in-circuit testers won't be economical, so dedicated testers have to be designed. Sometimes dedicated testers are also needed for integrating the product. Normally, smaller EMS partners will not have this skill and OEMs have to plan this to augment. This issue is one of the biggest trouble spots.
Design Validation	It is a key aspect, especially if the product happens to be a medical or an avionics device. Sometimes it is needed for telecom and NW products too. This capability is a value-added skill that helps.
Compliance Certification	OEMs get their product certified for standards, but when the volume of production increases process variations tend to impact the compliance, which should be avoided.
Reengineering & Value Engineering	It is predominantly targeted towards reduction of the product cost but needs a very close interaction with the EMS vendor's supply chain for a success.
Life Cycle Support (HW&SW)	This process allows the extension of the product life by addressing obsolescence issues.

theirs. Unfortunately, most of the times it is the other way round! One classic example is of an RF product manufactured by an OEM where the process of tuning needed intimate knowledge of the system, so complete outsourcing was not done. Instead, they did the final assembly in-house rather than outsourcing due to the complexity involved, and the product was a success.

2. Manufacturing cost alone cannot be the driver for outsourcing. Costs of transport, higher inventory levels due to distance and lead times, parts to be stocked for warranty, test and repair have to be factored in deciding the EMS partner.

3. This is critical if the product

is for a global market. As even the shipment packaging has to meet stringent standards. A well-planned product design and outsourcing strategy helps to a great extent.

4. Mature products with predictable schedules are the best candidates for outsourcing most of the times.

5. However, OEMs have to be aware of some of the risks like quality of components, impact of volume variations, and confidentiality when outsourcing their product for production. While most EMS partners adhere to rules, mistakes do happen due to business pressures.

6. Also, on the commercial side, OEMs need to factor in the local tax structure as mostly manufacturing

units enjoy some kind of incentives; unless OEMs understand that, EMS partners may not share them.

7. Financial stability and credit practices of a EMS partner is paramount, as when an EMS partner gets into financial trouble the inventories get locked up and OEMs' business suffers due to the sudden event.

What to look for when selecting an EMS partner

As mentioned earlier, choosing an EMS partner is not a transaction that you can do once. OEMs have to continuously spend time, energy, and money to maintain the relationship. The best approach is to select an EMS partner with whom you can